



ALL



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Complete the blanks in the following question with the appropriate answer.

Two jars, W and B, contain W_n and B_n balls respectively. W contains all white balls and B contains all black balls. You may perform the following actions. You may choose to perform one or both actions, or ignore both actions. Please note that the actions you perform must eventually result in maximizing the selection of a white ball when you are asked to select a ball from a selected jar further in the question.

Actions:

- a. Move any number of balls from W to B
- b. Move any number of balls from B to W.

After you have performed the above actions (that is if you chose to perform any action), you are blindfolded and asked to choose a jar at random and pick a ball from the chosen jar.

Note: You can enter the answer (probability) either with or without a leading 0. For example if the final answer is 0.25, you can enter either 0.25 or .25

W_n	45
B_n	56



ALL



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Actions:

- Move any number of balls from W to B
- Move any number of balls from B to W.

After you have performed the above actions (that is if you chose to perform any action), you are blindfolded and asked to choose a jar at random and pick a ball from the chosen jar.

Note: You can enter the answer (probability) either with or without a leading 0. For example if the final answer is 0.25, you can enter either 0.25 or .25

Wn	45
Bn	56

The maximum probability of choosing a white ball rounded to two decimal places is _____



ALL



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11. Question 11

Complete the blanks in the following question with the appropriate answer.

On a plane, there are M horizontal and N vertical lines. The distance between any two adjacent parallel lines is same. The vertical and horizontal lines intersect each other.

M	5
N	9

The total number of squares formed by these intersecting lines is 0



10. Question 10

ALL



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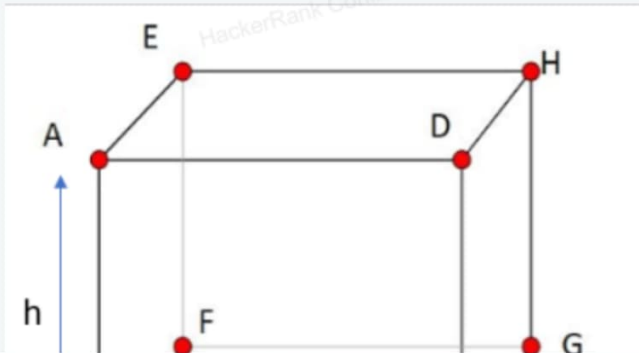
✓

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12

Complete the blanks in the following question with the appropriate answer.

In the given figure of a cuboidal room, we connect vertices A and G using a connecting wire of length y . The connecting wire always runs through the walls, floor or the roof, i.e. the wire is not visible externally. The dimensions of the room is $l \times b \times h$.





ALL



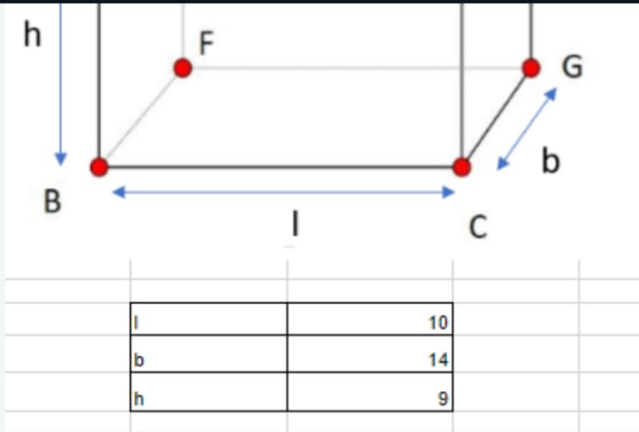
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12



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The minimum value of $y*y$ is 377.



ALL



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12. Question 12

Complete the blanks in the following question with the appropriate answer.

In a right triangle ABC, right angled at A, the median AM has a length of m . $AB:AC=k$.

k	7
m	10

The area of triangle ABC is 7.

Note: Round off the answer to first decimal place.



ALL



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13. Question 13

Complete the blanks in the following question with the appropriate answer.

a, b and c are natural numbers $(1, 2, 3, \dots)$ such that $ab+ca=p$ and $ca+bc=q$.

p	13
q	24

The least value of $ax+bx$ is 12



ALL



12



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16

A natural disaster was predicted by a government. To safeguard its citizens against the foreseen disaster, it started to migrate all the people from town A to town B.

As part of this process, all the vehicles in the town A are pressed into service, each one carrying the same number of people. Halfway towards town B, N number of vehicles broke down, so it was necessary for each of the remaining vehicles to carry one more person. Further on the journey towards town B, additional M number of vehicles broke down, which resulted in P more number of persons to be accommodated in each vehicle compared to what it had been when they started from town A.

N	10
M	15
P	7

Number of people in town A were _____ and the number of vehicles started from town A were _____



ALL



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15. Question 15

Complete the blanks in the following question with the appropriate answer.

A teacher handed over a paper bundle to a student and requested to count the number of sheets in the bundle. To count the number of sheets, the student started to mark the sheet numbers from 1 without any leading 0s. The number of decimal digits used to complete the sheet markings are N which is as given.

N	1575
---	------

The number of sheets that are present in the entire bundle was 15750



ALL



16

17

16. Question 16

Complete the blanks in the following question with the appropriate answer.

A couple was honored with best couple award this year and the present age of husband is double of their son's age + N years. Their son was born 10 years after their marriage and the husband's age at the time of marriage was $\frac{1}{3}$ rd of his present age.

N	1
---	---

Age of husband at the time of marriage was _____



ALL



16

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18

17. Question 17

Complete the blanks in the following question with the appropriate answer.

A black-box contains N pairs of shoes of M different colors. The contents of the black-box are all well shuffled, and it is not possible to look at any shoe without taking it out of the box.

N	76
M	8

You have been blindfolded and asked to take out one shoe at a time. Your job is to take out as few shoes as possible, yet make sure that you have got at least one matching pair to wear for yourself. Minimum number of shoes you need to take out is _____



ALL



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3

4

5

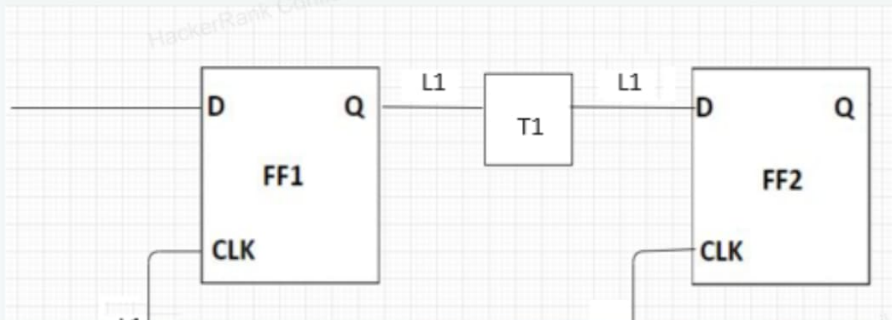
2. Question 2

Complete the blanks in the following question with the appropriate answer.

In the diagram, T_1/T_2 are the gate delays, L_1/L_2 are the wire lengths, where wire delay per unit length is T

Find the minimum clock period that is required for operating the circuit. Please provide the answer in ns.

Note - Setup Time, Hold time & Clock-to-q delays are the same for FF1 & FF2.





ALL

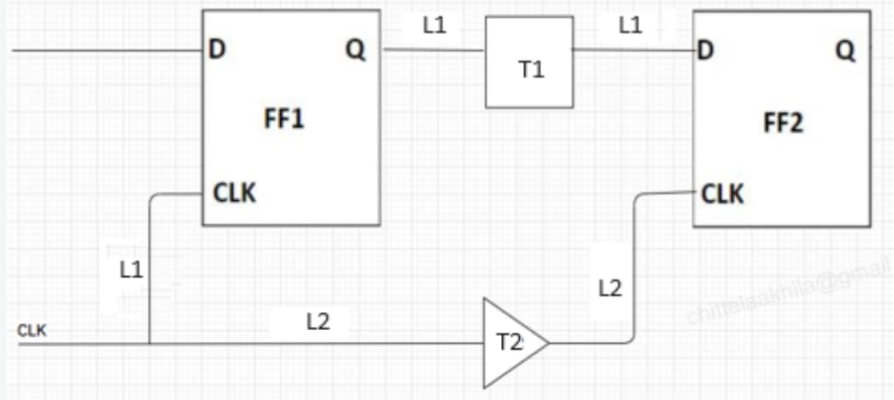


2

3

4

5



Wire Lengths

L1	2
L2	4



ALL



2

3

4

5

Timing Parameters

	min delay (ns)	max delay (ns)
T1	3	5
T2	2	4
T	2	6
tck to q	9	11
tsetup	4	
thold	2	

Min clock period the circuit can operate is 32 ns



The following block diagram shows an implementation of a computational logic. The area occupied by different components are shown below.

Redesign the block diagram to achieve the same functionality with the minimum possible area.

What is the total area of your design after minimization?

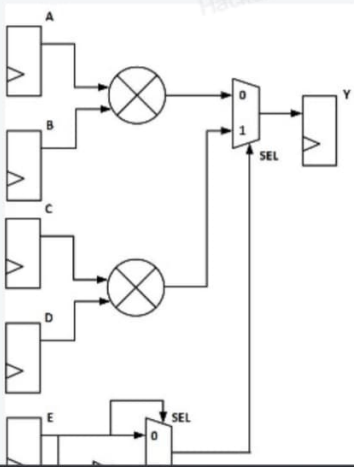
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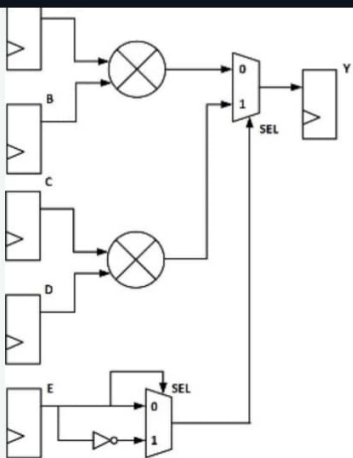
ALL



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Parameter	Unit	Value
Multiplier Area	sq. microns	1500



ALL



3

4



Parameter	Unit	Value
Multiplier Area	sq. microns	1500
Mux Area	sq. microns	50
Inverter Area	sq. microns	25
Register Area	sq. microns	50

Total area of the design after minimization is 1700 (sq. microns)



ALL



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5. Question 5

Complete the blanks in the following question with the appropriate answer.

Below image is an example of 16x4_M4 memory.

It has 16 addresses, 4 data bits and 4:1 Column Muxing.

This memory design has

4 Address bits,

4 Data bits

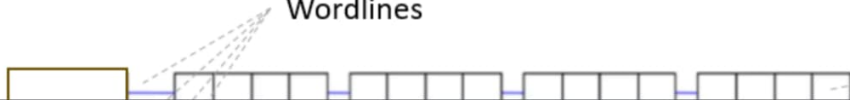
4 Wordlines,

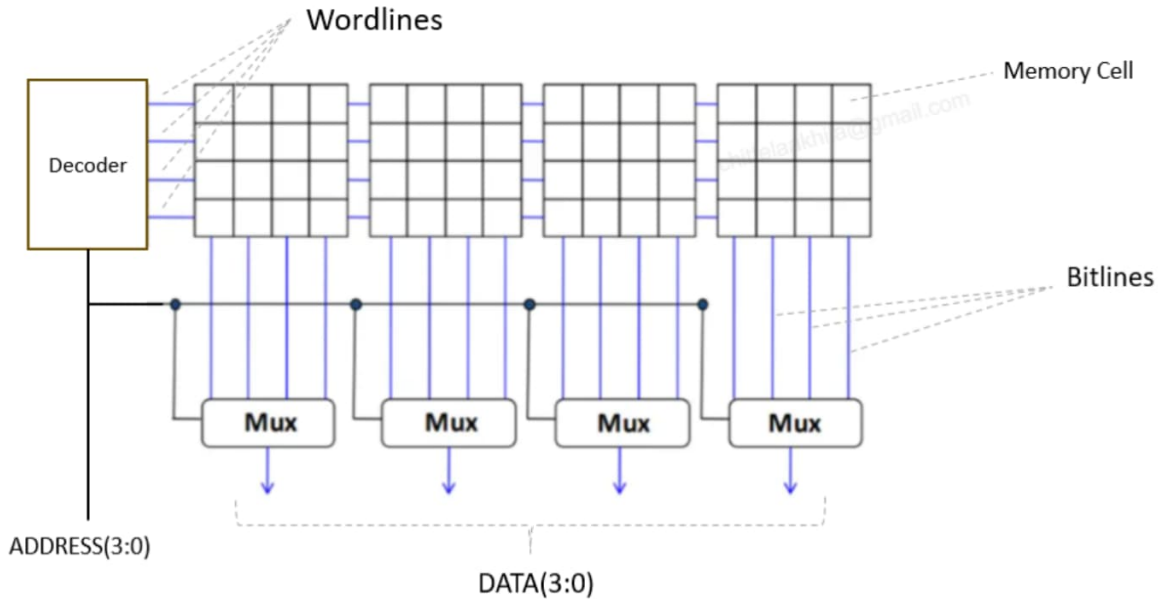
16 Bitlines,

64 Total memory cells

Wordlines

Memory Cell







ALL

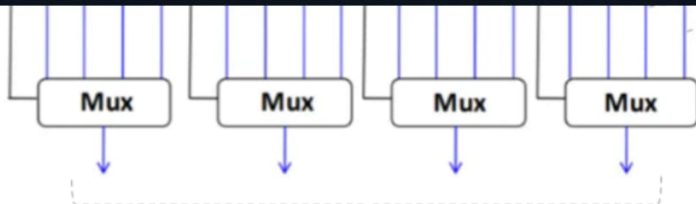


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ADDRESS(3:0)



DATA(3:0)

A 32x4_M4 memory design would have

a. 16 Wordlines

b. 32 Bitlines



ALL



6

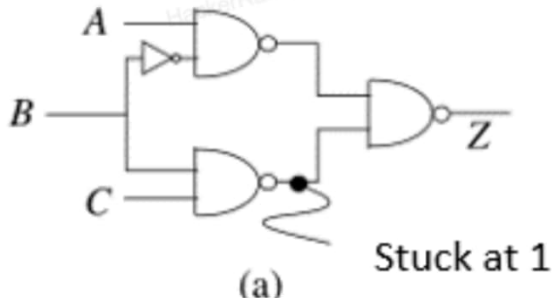
7

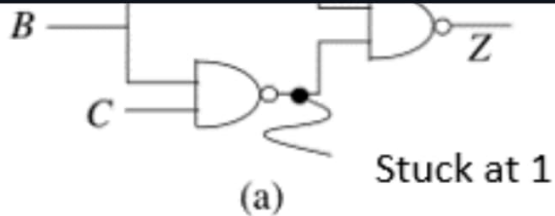
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6. Question 6

Complete the blanks in the following question with the appropriate answer.

A stuck at fault is a manufacturing defect where a node is permanently held at logical '1' or '0' or 'X'. Consider the circuits shown in Figure below with one or more faults





Note :

1. Use ' to represent complement value. For eg: If answer is NOT of X, then please write it as X'
 2. Use logical operators "|" and "&" for OR and AND resp.
 3. Use parentheses "(" ")" to group symbols.
 4. Don't use any spaces between the variables or logical operators.
- For eg : If answer is "(M AND N) OR (NOT of P)", then please write it as (M&N)|P'

Simplified logical expression for the output ("Z") in terms of the input variables ("A", "B", "C") for the given faults is $Z = A \& B'$



ALL



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Loading

8. Question 8

Complete the blanks in the following question with the appropriate answer.

A cache consists of 4 pages whereas the main memory consists of 16 pages. "Least Recently Used" policy is used for page replacement.

Initially, the cache contains the pages 1, 2, 3, 4. The Page access sequence in a program is as below

Sequence	3,2,1,5,4,3,2,1,3,6,7,2
----------	-------------------------

The Number of Page fault is 3

Submit Answer & Continue

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ALL



Complete the blanks in the following question with the appropriate answer.

What would be the output of this program given that data.inc contains the following information.

//data.inc

F(92,39)

```
#include <iostream>
#include <utility>

int main() {
    auto a =
        #define F(x,y) std::make_pair(x+y, x);
        #include "data.inc"
        #undef F
    auto b =
        #define F(x,y) std::make_pair(x##y, #x);
        #include "data.inc"
        #undef F

    std::cout << "First: " << a.first + b.first << std::endl;
```



```
#include <utility>
```

```
int main() {
```

```
    auto a =
```

```
        #define F(x,y) std::make_pair(x+y, x);
```

```
        #include "data.inc"
```

```
        #undef F
```

```
    auto b =
```

```
        #define F(x,y) std::make_pair(x##y, #x);
```

```
        #include "data.inc"
```

```
        #undef F
```

```
    std::cout << "First: " << a.first + b.first << std::endl;
```

```
    std::cout << "Second: " << a.second + 1 << b.second + 1 << std::endl;
```

```
}
```

First: _____

Second: _____